

Lab Sheet 10

Title: Work with Data Import and Aggregation Pipeline.

Objective:

- Import the Dataset to MongoDB.
- Working with Aggregation Framework.

Setup Instructions (Pre-requisite):

- Download and Install MongoDB Tools: <https://www.mongodb.com/try/download/database-tools>
- Download the appropriate version for your OS and install it.

Set Up Environment Variable (Path):

- Go to: System Properties → Environment Variables → Under System Variables → Select Path → Click Edit → Add the path where MongoDB tools (mongoimport) are installed.

C:\Program Files\MongoDB\Database Tools\100\bin

Verify Installation:

- Open CMD and run:
mongoimport --version

Importing Dataset into MongoDB

- Importing a JSON File:

cd C:\Path\To\Your\MongoDB\bin

mongoimport --jsonArray --db mydb --collection mycollection --file cars.json

Exercise:

- Import the dataset into a MongoDB database named CarSalesDB, collection cars.

```
mongoimport --jsonArray --db CarSalesDB --collection cars --file carsales.json
```

- Verify the creation of the database and collection using the following MongoDB shell (mongosh) commands

```
Show dbs  
use CarSalesDB  
Show Collections  
db.cars.find().pretty()
```

```
# Import JSON dataset into MongoDB
mongoimport --jsonArray --db CarSalesDB --collection cars --file
carsales.json

// Verify Database and Collection
show dbs

use CarSalesDB

show collections

db.cars.find().pretty()
```

- a. Find all cars sold by brand Toyota.

```
db.cars.find({
  brand: "Toyota"
})
```

- b. Find all cars with a price greater than 30000.

```
db.cars.find({
  price: { $gt: 30000 }
})
```

- c. Return the car_id, brand, and price of all cars sorted by price descending.

```
db.cars.find(
  {},
  {
    _id: 0,
    car_id: 1,
    brand: 1,
    price: 1
  }
).sort({ price: -1 })
```

- d. Find all Ford cars sold after 2024-07-01.

```
db.cars.find({
  $and: [
    { brand: "Ford" },
    { sold_date: { $gt: ("2024-07-01") } }
  ]
})
```

- e. Count how many cars were sold by each brand. (\$group)

```
db.cars.aggregate([
{
  $group: {
    _id: "$brand",
    totalCarsSold: { $sum: 1 }
  }
}]
```

- f. Find the total revenue from car sales.

```
db.cars.aggregate([
{
  $group: {
    _id: null,
    totalRevenue: { $sum: "$price" }
  }
}]
```

- g. Project only the brand, model, and sold_date for all documents.

```
db.cars.aggregate([
{
  $project: {
    _id: 0,
    brand: 1,
    model: 1,
    sold_date: 1
  }
}]
```

- h. Use \$match and \$group to find the total price of Toyota cars sold.

```
db.cars.aggregate([
{
  $match: {
    brand: "Toyota"
  }
},
{
  $group: {
    _id: "$brand",
    totalToyotaSales: { $sum: "$price" }
  }
}]
```